Title	TOPSHIELD: IOT SMART HELMET FOR ACCIDENT PREVENTION, SPEEDAND PROXIMITY DETECTION FEATURES	
GROUP 3 Members	Name	Ratings
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PROBLEM STATEMENT CANVAS

CONTEXT

When does the problem occur?

- The problem of motorcycle accidents occurs frequently in high-traffic urban areas, specifically due to overspeeding and lack of proximity detection, evidenced by a 17.3% increase in MMDA motorcycle accidents in 2023

PROBLEM

What is the root cause of the problem?

- The root cause is that traditional helmets offer only passive protection and do not integrate the proactive, real-time safety measures needed to address the active causes of accidents like overspeeding and collisions.

ALTERNATIVES

What do customers do now to fix the problem?

- Customers currently rely on traditional, passive motorcycle helmets for impact protection and may use separate, handheld or handlebar-mounted apps for speed tracking and GPS

CUSTOMERS

Who has the problem most often?

- The problem most affects Daily Commuters and Logistics Personnel (motorcycle riders) in urban environments who are highly vulnerable to preventable crashes.

EMOTIONAL IMPACT

How does the customer feel?

Riders feel unsafe, exposed, and vulnerable due to the high risk of severe consequences (90-95% chance of severe consequences), which creates significant anxiety for them and their families.

QUANTIFIABLE IMPACT

What is the measurable impact (include units)?

- The Metropolitan Manila Development Authority (MMDA) recorded

- 26,599 motorcycle accidents in 2023, averaging 78 accidents daily, contributing significantly to morbidity and fatality in the Philippines

ALTERNATIVE SHORTCOMINGS

What are the disadvantages of the alternatives?

- Alternatives fail because traditional helmets are not proactive, and using separate handheld devices is

- prohibited by law (Anti-Distracted Driving Act) and creates dangerous visual distraction, lacking the necessary integrated, auditory warnings.